



TavaOne Education

General License Course · 2023–2027 Pool

Module 2

Operating & Propagation

General License Course · G2 + G3 · 8 Exam Questions



Hour 1

HF Operating Practices



G2A — SSB Operating: The Basics

SSB is THE mode for HF phone — AM is rarely used (wastes power and bandwidth)

LSB (Lower Sideband) below 10 MHz: 80m, 40m use LSB by convention

USB (Upper Sideband) above 10 MHz: 20m, 17m, 15m, 12m, 10m use USB

Exception: 60m always uses USB regardless of frequency (channelized operation)

Before transmitting: listen first! Check if the frequency is in use

"QRL?" = "Is this frequency in use?" — the polite first check

If calling CQ: "CQ CQ CQ, this is W4GGJ, Whiskey Four Golf Golf Juliet, CQ and standing by"

DX station: send your call once or twice — don't monologue at the DX



G2A — Calling Frequencies by Band

80m CW calling: 3.585 MHz (approximate — varies by region)

80m/75m SSB calling: 3.940, 3.960, 3.975 MHz (regional nets vary)

40m SSB: 7.200, 7.250 MHz (General phone subband starts 7.175)

20m SSB calling: 14.225 MHz (also 14.230 SSTV, 14.300 maritime mobile)

15m: 21.300 MHz

10m SSB: 28.400 MHz; FM calling: 29.600 MHz

These are band plans (voluntary) not FCC allocations (mandatory)

Band plans tell you where people actually hang out — FCC only tells you which frequencies you're allowed to use



G2B — Operating Courtesy

Check if a frequency is in use BEFORE calling CQ — listen for a minute

"QRL?" before transmitting; wait 5–10 seconds for a response

Do NOT transmit on top of an ongoing contact — find another frequency

DX pileup operation: send your call once, wait and listen; don't repeat over and over

Working split: DX listens on one frequency, transmits on another — you transmit on the DX's listening frequency

Priority: distress (MAYDAY) > emergency traffic > other traffic

Band plans specify calling frequencies — don't rag-chew on a calling frequency (call CQ, then QSY)

Avoid operating near the band edge — dial drift and SSB sidebands extend beyond your carrier



G2C — Q Signals You Need to Know

During a QSO

QRL — Is this frequency in use? / This frequency is in use

QRN — Static (natural noise)

QRM — Man-made interference

QSB — Fading (signal varies)

QRZ — Who is calling me?

QSL — I acknowledge / I confirm

QTH — My location is...

QSY — Change to another frequency

General Operating

QRP — Low power (5W or less)

QRX — Wait / I will call you again

QRN — Troublesome static

QRO — Increase power (or high power)

QRS — Send more slowly

QRQ — Send faster

73 — Best regards

SK — End of contact (end work)



G2D — ARES and RACES

ARES (Amateur Radio Emergency Service): ARRL-affiliated volunteer organization

ARES provides emergency communications during disasters — not government-controlled

Any licensed amateur can join ARES — no special FCC registration required

RACES (Radio Amateur Civil Emergency Service): government-affiliated, activated by civil defense

RACES requires FCC registration of the station AND enrollment in the civil defense organization

Only RACES personnel may operate RACES nets when activated — non-RACES amateurs may not participate

Key distinction: ARES = ARRL volunteer, activate on your own; RACES = government-activated, requires registration

In practice: many hams are dual-registered in both organizations



G2E — Digital Modes on HF

RTTY: radioteletype — the oldest digital mode still in use; 170 Hz shift, 45.45 baud; uses AFSK on LSB

PSK31: phase-shift keying at 31 baud; ~80 Hz wide; keyboard-to-keyboard; dial + audio offset = RF frequency

FT8: 15-second transmit/receive cycles; requires clock sync within ± 1 second; 50 Hz bandwidth; can decode -20 dB signals

Winlink: email over radio; uses HF gateways; PACTOR or Ardop modems

APRS: Automatic Packet Reporting System; 1200 baud on VHF; 300 baud on HF; digipeaters relay packets

Bandwidth matters: choose a mode whose bandwidth fits within your subband allocation

PSK31 note: your dial frequency is NOT your RF frequency — add the audio tone frequency to find actual RF

FT8: you exchange grid squares and signal reports; full QSO is automated; uses WSJT-X software



Hour 2

The Ionosphere and HF Propagation



G3A — The Ionosphere: Your HF Backboard

The ionosphere: layers of ionized gas 60–600 km up; ionized by solar UV and X-ray radiation

D LAYER (60–90 km): lowest; absorbs signals, especially at LF/MF/low HF; exists only during daylight; disappears at night
→ why 80m/40m open up after dark

E LAYER (90–140 km): supports sporadic-E (Es); unpredictable; can open 10m/6m suddenly; seasonal (summer solstice peak)

F1 LAYER (140–200 km): daytime only; merges with F2 at night

F2 LAYER (200–600 km): the DX layer; reflects signals thousands of miles; higher during solar maximum

F1+F2 merge at night into single F layer → night brings better DX on 40m/20m due to D layer disappearing

Solar maximum: more ionization → F2 layer stronger → higher bands (15m, 10m, 12m) work reliably

Solar minimum: F2 weak → upper bands often dead; 20m and lower still work



G3B — MUF, LUF, and Skip

Skip: radio wave goes up at an angle, reflects off the ionosphere, and lands at a distance — the "skip distance"

Skip zone: the area between your ground wave range and where the skip lands — nobody there hears you

MUF (Maximum Usable Frequency): highest frequency that will reflect off the ionosphere for a given path

Stay below the MUF — frequencies above it pass through and go to space

LUF (Lowest Usable Frequency): lowest frequency usable before D-layer absorption kills the signal

Your working frequency must be between LUF and MUF to get through

As MUF decreases (toward night or solar minimum): switch to a lower-frequency band

Gray line: the terminator between day and night; enhanced propagation; D layer is thin → less absorption



G3C — Propagation by Band

Lower HF (160m, 80m, 40m)

160m/80m: mostly nighttime; D layer absorbs during day; reliable regional

40m: workhorse band; nighttime DX, daytime regional

NVIS (Near Vertical Incidence Skywave): 80m/40m; antenna pointing nearly straight up; covers 100–500 miles; used by emergency nets

Key: these bands are worst during peak sunlight

Upper HF (20m, 17m, 15m, 12m, 10m)

20m: most reliable DX band; works most of solar cycle

15m/12m: excellent near solar maximum; often closes near minimum

10m: spectacular when open; long dead periods near solar minimum

6m: the "magic band" — sporadic E openings; EME; meteor scatter

Key: these bands need the F2 layer strong



Solar Indices: SFI, A-Index, K-Index

Solar Flux Index (SFI): measures solar radio emission at 10.7cm; higher = better F-layer ionization

SFI >150: excellent conditions on 10m, 15m, 20m; expect DX

SFI <80: 10m usually dead; 15m marginal; focus on 20m and lower

A-Index: daily average geomagnetic activity (0=calm, >50=major storm); lower is better for HF

K-Index: 3-hour geomagnetic activity (0–9 scale); lower is better; updated 8 times/day

High K-index = geomagnetic storm = poor HF propagation, especially on polar paths

Best HF conditions: high SFI, low A-index, low K-index

Real-time data: DX Maps (dxmaps.com), PSKreporter, solar.tavaoneeducation.org



Checkpoint Quiz

Which sideband is commonly used for HF phone operation above 10 MHz?

A. Upper sideband

B. Lower sideband

C. Double sideband



Checkpoint Quiz

What is the Q signal for "Is this frequency in use?"

A. QRL

B. QRM

C. QRN



Checkpoint Quiz

Which ionospheric layer is responsible for long-distance HF communication?

A. F2 layer

B. D layer

C. E layer



Checkpoint Quiz

What does the term "Maximum Usable Frequency" (MUF) describe?

A. The highest frequency that will be refracted back to Earth by the ionosphere on a given path

B. The highest frequency the transceiver can generate

C. The maximum frequency permitted by FCC on a given band



Checkpoint Quiz

What is the primary difference between ARES and RACES?

A. RACES is activated by a civil defense organization; ARES is ARRL-affiliated and can activate independently

B. ARES requires government registration; RACES does not

C. RACES operates on VHF only; ARES operates on all bands



HamStudy Assignment

Drill these sections before next module:

G2A — Phone & CW Operating

G2B — Operating Courtesy

G2C — CW Procedures

G2D — ARES & RACES

G2E — Digital Modes

G3A — Ionosphere Layers

G3B — MUF/LUF/Skip

G3C — Band Propagation

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